BUNYAN YORGANCI

COMPUTER SYSTEMS ENGINEERING | EMBEDDED SOFTWARE & SYSTEMS | C / C++ / PYTHON

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EDUCATION

Carleton University | *B.Eng., Computer Systems Engineering* Expected Apr 2029

Selected coursework: Operating Systems, Computer Architecture, Real-Time Systems, Digital Systems, Communications, Circuits & Signals

TECHNICAL SKILLS

Languages: C, C++, Python, Java; MATLAB, Bash / Unix CLI

Embedded / hardware: Raspberry Pi, Arduino, GPIO, PWM, I²C, Sense HAT, motor drivers, circuit troubleshooting, soldering

Development / testing: Linux, Git, JUnit 5, Gradle, JaCoCo, Flask, Socket.IO, OpenCV, SQLite, MATLAB/Simulink, Logisim

SELECTED ENGINEERING PROJECTS

Operating-System & Interrupt Simulator | *C++, Linux* | *SYSC 4001* 2024–25

- Implemented trace-driven CPU event, interrupt and system-call simulation with process-control-block state transitions.
- Modeled FORK/EXEC, fixed-partition memory allocation and scheduling; analyzed execution traces and I/O scenarios.

Linux Process Synchronization | *C, POSIX IPC* | *SYSC 4001* 2025

- Implemented multiprocess coursework code using fork(), shared memory and semaphores for coordinated access.
- Investigated contention and synchronization behavior using test outputs, including a documented livelock case.

Raspberry Pi Hardware Control & Monitoring | *Python, GPIO, Flask* | *SYSC 3010* 2025

- Programmed traffic-light timing and pedestrian-button behavior; used assertions to verify hardware control states.
- Built browser-to-Sense-HAT LED controls with Flask-SocketIO; completed separate OpenCV monitoring and SQLite logging labs.

Motor & Display Hardware Interfaces | *Python, GPIO, PWM, I²C* | *ECOR 1044* 2022

- Completed Raspberry Pi mechatronics labs on motor direction, stepper sequencing, servo positioning and LCD updates.
- Used PWM for motor control, an eight-step coil sequence for a stepper motor and I²C commands for LCD output.

Software Validation & Boundary Testing | *Java, JUnit 5, Gradle, JaCoCo* | *SYSC 4101* 2026

- Designed parameterized JUnit boundary tests reaching 100% instruction and branch coverage for Board.withinBorders() only.
- Covered valid positions and each out-of-range direction; generated JaCoCo reports through Gradle.

Mobile Robot Motor Control — In Progress | *Arduino, C/C++* | *SYSC 4805* 2026

- Developing motor-driver direction and stop routines for a four-motor platform; rear encoder integration is in progress.
- Organizing the motor control interface into forward, reverse, turning and stop operations for Arduino-based testing.

Digital Logic & Seven-Segment Decoder | *Logisim* | *SYSC 2310* 2022

- Designed gate-level combinational logic and tested four-bit inputs against seven-segment display outputs in simulation.

ADDITIONAL TECHNICAL WORK & EXPERIENCE

Communications simulation: Modeled BPSK/QPSK modulation, pulse shaping and channel/SNR effects in course labs (SYSC 3501).

Systems simulation: Modeled dynamic systems using MATLAB and Simulink (SYSC 3600).

Computer refurbishment: Self-employed (2017–present): diagnosed hardware faults, replaced computer components and restored operating systems.